

NGAME Deployment Plan

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User

Need to leave the Intuit Developer environment and access a customer’s actual QBO account via OAuth. I won’t have the Developer’s prompts in order to get the necessary OAuth id’s, secretts, and the rest. How do I set up that OAuth process?

Cursor

Let me look at the project context first to understand what integration is already in place.

Good news: your project already has a solid OAuth implementation in `ngame_quickbooks_oauth.py` . Here’s what you need to understand about the full picture, because there’s a common misconception worth clearing up first.

The Fundamental Model

You **always** need an Intuit Developer account — but it’s **yours**, not the customer’s. Here’s the separation of concerns:

Who	What they provide
You (the developer)	<code>client_id</code> and <code>client_secret</code> — from your registered app in the Intuit Developer Portal
The customer	Authorization — they click “Connect to QuickBooks” and grant your app access to their company (<code>realm_id</code>)

The OAuth flow exchanges those two things for tokens your app uses to read the customer’s data. There is no path that skips the developer portal; the `client_id` / `client_secret` pair is what proves to Intuit that your application is legitimate.

Step-by-Step Setup for a Real Customer Account

1. Create / configure your Intuit app for Production

1. Go to developer.intuit.com → **My Apps** → **Create an App** → **QuickBooks Online and Payments**.
2. Under **Keys & OAuth**, you'll see both **Sandbox** and **Production** keys. Copy the **Production** Client ID and Client Secret .
3. Under **Redirect URIs**, add exactly the URI your local server will listen on. Your code defaults to `http://localhost:8000/callback` — add that (or whatever port you prefer). Intuit requires this to match exactly.

2. Update `quickbooks_config.json`

Change environment from "sandbox" to "production" and paste your production credentials:

```
{
  "quickbooks_api": {
    "client_id": "YOUR_PRODUCTION_CLIENT_ID",
    "client_secret": "YOUR_PRODUCTION_CLIENT_SECRET",
    "access_token": "",
    "refresh_token": "",
    "realm_id": "",
    "environment": "production",
    "redirect_uri": "http://localhost:8000/callback"
  }
}
```

Or, preferably (to keep secrets out of the file), set environment variables and leave the file token-only:

```
export QBO_CLIENT_ID="YOUR_PRODUCTION_CLIENT_ID"
export QBO_CLIENT_SECRET="YOUR_PRODUCTION_CLIENT_SECRET"
export QBO_ENVIRONMENT="production"
```

3. Run the interactive OAuth flow once per customer

Your existing `ensure_quickbooks_auth()` function already handles this automatically. When `access_token` and `refresh_token` are blank, it falls through to the interactive path (lines 266–306 in `ngame_quickbooks_oauth.py`):

1. It opens a browser to Intuit’s authorization page.
2. The **customer** signs into their QBO account and clicks **Authorize**.
3. Intuit redirects back to localhost:8000/callback with a code and realmId .
4. Your local server captures them, exchanges the code for tokens, and writes everything back to quickbooks_config.json .

From that point on, the refresh token handles re-authorization silently until it expires (~100 days of inactivity) or the customer revokes access.

4. Trigger that first-time flow

The simplest way is to call ensure_quickbooks_auth directly from the command line:

```
# one-time bootstrap script (save as bootstrap_oauth.py)
from ngame_quickbooks_oauth import ensure_quickbooks_auth, default_quickbooks_con

bundle = ensure_quickbooks_auth(default_quickbooks_config_path())
print(f"Authorized. Realm ID: {bundle.realm_id}")
```



Run it with:

```
python bootstrap_oauth.py
```

The browser will open, the customer authorizes, and quickbooks_config.json will be populated with live production tokens.

Key Differences: Sandbox vs. Production

Setting	Sandbox	Production
environment	"sandbox"	"production"
client_id / client_secret	From Development keys tab	From Production keys tab
Customer’s data	Intuit test data only	Real customer QBO company

Setting	Sandbox	Production
Redirect URI	Must be registered in portal	Must be registered in portal
realm_id	Intuit-generated test company ID	Customer's actual company ID

What You Do NOT Need

- You do **not** need the customer's Intuit developer credentials.
- You do **not** need anything from inside their QBO account settings.
- You do **not** need to touch the Intuit portal again after initial app setup — the OAuth flow is entirely self-contained once your `client_id` / `client_secret` are in place.

The customer just needs to be the QBO company **owner or admin** (to grant authorization) and have internet access to complete the browser-based consent screen.

User

Clarify and correct my misconception or over-expectation. For the real go-live, I want NGAME to run on a customer's "Surveillance" computer, not mine. I want for the customer, using their own Surveillance computer, to access their own (the customer's) QBO from the bookkeeper's machine.. The customer should not be an Intuit Developer. All of NGAME must install and run on the Surveillance machine. That machine belongs to the financially responsible person (FRP). That person runs the training phase, watches his dashboard for warnings, and runs any further enhancements. The FRP is not a developer and should not have to subscribe to Intuit's developer program. I.E. The NGAME "customer" is the FRP.

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This is an important architectural clarification, and your mental model is actually correct for the end goal. The misconception is smaller than you think. Let me lay it out clearly.

The Misconception: Who Registers the App

You are right that the FRP should never touch the Intuit Developer Portal. But **someone** must register NGAME as an application with Intuit — exactly once, ever — and that person is **you (Ron)**, not the customer.

This is identical to how every QBO integration in the world works:

- **Gusto** (payroll) registered their app with Intuit. Their customers never become Intuit developers. Customers just click “Connect to QuickBooks.”
- **Bill.com** did the same.
- **NGAME** will do the same.

The `client_id` and `client_secret` are **NGAME’s identity as a software product** — not the customer’s credentials. You register the app once, you bundle those credentials into the software you distribute, and you never expose them to the FRP.

The Two-Layer Credential Model

Layer	What it is	Who owns it	Where it lives
App credentials	<code>client_id</code> , <code>client_secret</code>	You (Ron / NGAME LLC)	Bundled into the NGAME installation — FRP never sees these
Customer tokens	<code>access_token</code> , <code>refresh_token</code> , <code>realm_id</code>	The FRP’s QBO company	Written to <code>quickbooks_config.json</code> on the Surveillance computer after one-time consent

The FRP’s Actual Experience

1. Install NGAME on the Surveillance computer (one installer, no developer setup).
2. Run NGAME for the first time.
3. A browser opens: **“NGAME wants to access your QuickBooks company. Sign in to authorize.”**
4. The FRP logs into their own QBO account (the same one the bookkeeper uses) and clicks **Connect**.
5. Browser says “NGAME: OAuth complete. You can close this tab.”

6. Done. NGAME now has a refresh token that lasts ~100 days of inactivity and silently renews on every run.

The FRP never touches the Intuit Developer Portal. They never see a `client_id`. They just log into their own QBO.

What This Requires You to Do (Once)

1. **Register NGAME as a production app** at developer.intuit.com.
2. Get your production `client_id` and `client_secret`.
3. **Bundle those into the NGAME distribution** — not in `quickbooks_config.json` (which is customer-writable), but in a separate read-only location (environment variable injected at install time, a protected config file, or compiled in). The current code already supports `QBO_CLIENT_ID` / `QBO_CLIENT_SECRET` as env vars, which is the right hook for this.
4. Ship it.

Note on Intuit's review process: For production access to real QBO companies, Intuit requires an app review before you can serve more than ~25 companies. For a controlled rollout to a handful of FRP customers, you can operate under the "development" tier of a production app while the formal review is pending.

What Needs to Change in the Current Codebase

Right now `quickbooks_config.example.json` puts `client_id` and `client_secret` alongside the customer tokens — implying the FRP fills them in. That needs to be restructured so the FRP's config only contains what belongs to them:

```
{
  "quickbooks_api": {
    "access_token": "",
    "refresh_token": "",
    "realm_id": "",
    "environment": "production",
    "redirect_uri": "http://localhost:8000/callback"
  }
}
```

And the app credentials arrive via the installer (as `QBO_CLIENT_ID` / `QBO_CLIENT_SECRET` in the shell environment, a `.env` file the installer writes, or an OS keychain entry). Your

ngame_quickbooks_oauth.py already reads those env vars and already strips them from the JSON it writes back — so the plumbing is already correct. Only the documentation and example config need to reflect this intent.

Summary

What you thought was needed	Reality
FRP needs an Intuit Developer account	No — you register NGAME as an app once
FRP fills in <code>client_id</code> / <code>client_secret</code>	No — those are bundled in the NGAME installer by you
FRP does something technical with OAuth	No — they just click “Connect to QuickBooks” in a browser
FRP manages tokens	No — NGAME auto-refreshes silently on every run

Your deployment vision is sound. NGAME is a product; you are its publisher; the FRP is just an end user who consents to QBO access once. That is exactly how every commercial QBO integration works.

User

Good. That response is a well-compartmentalized deployment roadmap for the end product. Now fell back to my current development environment. I’m ready to field-test NGAME with a trusted and trusting organization, a small non-profit. They know that this is a testing period only, but they’re willing to trust me as a consultant and still as a developer to play the role of the FRP. For this scenario, I assume that there is very little I need to do or that this “test site” non-profit organization needs to do. I’ll appreciate the same table-style response that you’ve provided so far.

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Field Test with the Non-Profit: What’s Actually Required

The short answer: Almost nothing from the non-profit, a few small steps from you.

What the Non-Profit Needs to Do

Task	Details	Effort
Have an active QBO subscription	They almost certainly already do	Zero — pre-existing
Designate one QBO admin to authorize	The person who clicks "Connect" must have QBO Admin or Company Admin rights	2 minutes, one time
Be present (or hand you their login) for the OAuth consent screen	They log into their own QBO and click Authorize	2 minutes, one time

That is the complete list. No installs, no accounts, no technical knowledge required.

What You Need to Do

Task	How	One-time or recurring
Switch to your app's Production keys in the Intuit Developer Portal	Your existing app already has a Production tab alongside Sandbox — copy the Production Client ID and Client Secret	One-time
Update <code>quickbooks_config.json</code> on your machine	Change <code>"environment": "sandbox"</code> → <code>"production"</code> , paste production credentials	One-time per test site
Clear out the old sandbox tokens	Set <code>access_token</code> , <code>refresh_token</code> , and <code>realm_id</code> to <code>""</code> in the config	One-time
Register your redirect URI under the Production keys	In the Developer Portal → your app → Production → Redirect URIs — confirm	One-time

Task	How	One-time or recurring
	<code>http://localhost:8000/callback</code> is listed	
Run the OAuth flow	Run any NGAME script that calls <code>ensure_quickbooks_auth()</code> — your browser opens, the non-profit admin logs in and authorizes, tokens are saved automatically	One-time, then silent
Run NGAME normally	Training phase, dashboard, everything else — identical to your sandbox workflow	Ongoing

What Does NOT Change

Thing	Status
Your Intuit Developer account	No changes needed — you already have one
NGAME codebase	No code changes needed
Your machine as the “Surveillance computer”	You’re playing the FRP role — your machine is the surveillance computer for this test
The non-profit’s QBO workflow	Nothing changes for the bookkeeper — they keep doing what they do
Intuit app review / certification	Not required yet — production apps can serve up to 25 companies before Intuit’s formal review is required. One non-profit is well within that limit

The One Gotcha to Watch

Intuit’s OAuth consent screen will show your **app name** (whatever you named it in the Developer Portal) to the non-profit admin when they authorize. Make sure the app name is something recognizable and professional — e.g. “NGAME Financial Surveillance” — so they aren’t confused by a generic or placeholder name when they click Authorize.

User

OK. That scenario—testing with a live non-profit who is willing to support my development. New another scenario: The School of Business in a university wants to incorporate NGAME into their practice lab. They will use QBO to set up fictitious non-profit companies on their own and completely from scratch: Chart of Accounts, donors.... everything. They will get their own QBO license directly from Intuit. Assume that NGAME can now be deployed commercially, based on successful field tests, such as the one whose guidance you provided above (the trusted site scenario). My assumption is that a computer or a group of computers in that lab or a select group of students will each have an NGAME license and that NGAME willl be installed on those students' surveillance computers. Same request to you: proved the same format of tables, explanations, and warnings as you have thus far.

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University Business School Lab Deployment

Context: What Makes This Scenario Different

There are three meaningful differences from the field-test scenario that cascade through everything else:

Factor	Field Test	University Lab
Number of QBO companies	1	Many (one per student or team)
Who sets up QBO	Pre-existing organization	University builds from scratch
Intuit app scale	Under 25-company threshold	Likely exceeds it — full certification required
NGAME distribution	You install it manually	Packaged installer; students self-install
Your role	Developer acting as FRP	Software vendor; faculty/students are FRPs

What Ron Must Complete Before This Deployment

Task	Details	Timing
Pass Intuit's production app review	Required once you exceed ~25 connected companies. Submit through the Developer Portal → App Assessment. Intuit reviews security practices, data handling, and UI. Timeline is typically 2–6 weeks.	Before launch
Finalize bundled credentials	<code>client_id</code> and <code>client_secret</code> must be embedded in the installer — never exposed to students	Before installer is built
Build a packaged installer	Students should run one installer (e.g. a <code>.pkg</code> for Mac or <code>.exe</code> for Windows) that installs Python environment, NGAME, dependencies, and pre-fills app credentials	Before launch
Establish a NGAME license model	Per-seat, per-company, or per-institution — the university buys one institutional license and distributes to students	Before launch

What the University Needs to Do

Task	Details	Who Does It	Effort
Obtain QBO licenses	One QBO company file per student or team. Intuit has an academic/educational program — the university should contact Intuit directly to ask about volume academic pricing or the Intuit Education Program . This can significantly reduce per-seat cost.	Dept. administrator	One-time procurement
Build each fictitious company	Chart of Accounts, donor list, transaction history — this is the educational exercise itself. Each company should be set up to produce meaningful training data for NGAME.	Faculty / students	Ongoing (academic exercise)

Task	Details	Who Does It	Effort
Designate one QBO Admin per company	Each fictitious QBO company needs at least one user with Admin rights. For a student company, that student is the Admin.	Faculty	At company setup
Purchase an institutional NGAME license	One transaction with Ron; covers the full lab cohort	Dept. administrator	One-time procurement
Distribute NGAME installer to students	Standard IT distribution — shared drive, LMS download, or USB. No special configuration needed per student	IT / faculty	Per semester

What Each Student Needs to Do

Task	Details	Effort
Install NGAME on their surveillance computer	Run the installer Ron provides. App credentials are already bundled — nothing to configure manually.	5 minutes, one-time
Authorize NGAME against their QBO company	On first run, browser opens → student logs into their fictitious QBO company → clicks Authorize → done. Tokens save automatically.	2 minutes, one-time
Run the training phase	Same as any FRP — establish baseline behavior from their company’s transaction history	Per lab exercise
Monitor the dashboard	Watch for anomalies their classmates (playing the bookkeeper role) introduce	Ongoing during semester

The Bookkeeper / FRP Role Split in a Lab Context

This is worth calling out explicitly because the lab has a unique dynamic:

Role	Who plays it	Machine
Bookkeeper	Student (or a designated “fraud actor” student) enters transactions in QBO from any machine — QBO is cloud-based, so any browser works	Any lab computer or personal laptop
FRP / NGAME operator	Same student (or a different student pair) runs NGAME on the Surveillance computer	The designated surveillance computer

A common lab exercise structure: Student A plays the bookkeeper and introduces a hidden anomaly. Student B plays the FRP and tries to detect it using NGAME. Then they swap. Nothing about this requires any changes to NGAME.

Faculty Supervision Option (Future Enhancement to Flag)

A faculty member may want a single dashboard that monitors **all** student companies simultaneously — one pane of glass across the whole lab. NGAME currently runs one company per installation. This is a meaningful product enhancement to plan for the academic market: a **multi-tenant faculty dashboard** that connects to N companies with N separate OAuth tokens. Not required for launch, but worth noting as a near-term roadmap item.

Warnings and Gotchas

Risk	Details
Intuit app certification timeline	Plan 6–8 weeks minimum before the semester starts. Do not assume it will be fast.
QBO Academic Program availability	Intuit’s academic pricing has changed over the years. Confirm current terms directly with Intuit — do not promise the university a price point until you have it in writing.
Each company needs its own OAuth consent	There is no bulk authorization. Each student runs the consent flow once for their company. Faculty cannot pre-authorize on behalf of students.
Shared computers in a lab	If students share lab machines (not personal laptops), each <code>quickbooks_config.json</code> must be in a student-specific profile or

Risk	Details
	directory. Otherwise Student B will overwrite Student A's tokens. Build this into the installer's directory structure.
QBO company must have enough transaction history	NGAME's training phase needs sufficient data to establish a meaningful baseline. A brand-new fictitious company will need a few weeks of simulated transactions before training produces useful z-scores. Build this into the lab timeline.
Refresh token expiration	Tokens expire after ~100 days of inactivity. If a company goes unused between semesters, students will need to re-authorize at the start of the next term. This is a 2-minute task but faculty should budget for it.

User

Re population of a fictitious company's data, is it not possible to import directly a pre-built company via a QBB file, as with the old QB Desktop? I'm not a QBO expert. I only know what I did with QB Desktop in my former company. So this is a QB user's question.

Cursor

This is a QuickBooks product question, not a coding task, so no plan is needed — just a direct answer.

QBB Files and QBO: The Short Answer

No. QBO does not accept QBB files. QBB is a QB Desktop backup format. QBO is an entirely separate product with a different data model, and Intuit has never bridged the two in that direction.

What QBO Does Support for Pre-Populating a Company

Method	What it covers	Practical for a university lab?
CSV import	Chart of Accounts, Customers, Vendors, Products/Services	Yes — faculty can build template CSV files once and students import them at company setup. This is the closest equivalent to “loading a starter company.”
IIF import	Some transaction types (limited, legacy format)	Possible but finicky. Not recommended for a clean lab setup.
Third-party import tools (Transaction Pro, SaasAnt, etc.)	Full transaction history in bulk	Yes, but adds a paid tool and complexity. Overkill for a lab unless deep history is needed for NGAME training baseline.
Intuit’s Desktop-to-Online migration tool	Full company migration from a real QBB file	Technically yes, but it’s designed for one-time live migrations of real companies, not for cloning fictitious training companies. Intuit’s terms likely prohibit using it as a lab template mechanism.
QBO Sample Company (“Craig’s Design and Landscaping”)	A pre-built Intuit-provided practice company	Good for exploration, but students cannot modify or own it, and it cannot be connected to NGAME via OAuth since it isn’t a real company under the student’s account.
Manual entry	Everything	Always works; time-consuming; is itself the accounting lab exercise

The Practical Recommendation for the Lab

The most realistic approach for the university scenario is a **faculty-maintained set of CSV import templates**:

- One CSV for Chart of Accounts (non-profit structure: program funds, restricted/unrestricted net assets, donor revenue accounts, etc.)
- One CSV for Vendors and Donors (fictitious names)

- A seed set of historical transactions imported via CSV or a tool like SaasAnt

Students import those templates at company setup — takes about 15 minutes — and the company is immediately meaningful enough for NGAME’s training phase. Faculty maintains the templates as a versioned file that evolves with the curriculum.

This is worth flagging to the university’s faculty contact early, because “set up a company from scratch” underestimates the setup time if they’re expecting a QBB-style instant restore. The CSV template approach solves it neatly.

User

Very general marketing question for confirmation: NGAME is a self-contained application that accesses QBO accounts. The QBO never sees even a hint of NGAME in any screen. I maintain that NGAME is therefore not really a “QB Add-on” but a helper app. My intention is that students pursuing a career in auditing will nevertheless develop a reliance on NGAME and will expect to incorporate into their professional “toolbox”

Cursor

Your characterization is technically accurate and the marketing positioning is sound. A few confirmations and one nuance worth knowing.

NGAME’s Relationship to QBO: Technical Reality

Characteristic	Reality
Appears in QBO’s daily interface	No — the bookkeeper never sees any trace of NGAME
Appears in QBO’s App Store	No — NGAME is not listed there and does not need to be
Modifies QBO data in any way	No — NGAME is read-only. It queries; it never writes.
Runs inside QBO’s environment	No — it runs entirely on the Surveillance computer

Characteristic	Reality
Is it a "QB Add-on"?	No, by any reasonable definition. It is an independent application that happens to use QBO as its data source.

The one nuance: NGAME does appear in one obscure QBO location — the company’s **Settings → Connected Apps** list. A QBO admin looking in that settings screen would see that "NGAME" (your app name) has been granted read access. This is a standard OAuth disclosure, not an integration. The bookkeeper doing daily work sees nothing. This is worth knowing but does not affect your positioning.

The Right Analogy: ACL and IDEA

Your positioning maps precisely onto an already-established category in the auditing profession. Two well-known examples:

Tool	What it does	Relationship to the accounting system
ACL (Galvanize / Diligent)	Data analytics for auditors — extracts and analyzes GL data	External. Never touches the accounting system’s UI. Auditors consider it a core career skill.
IDEA (CaseWare)	Same category — fraud detection, transaction analysis	External. Taught in accounting programs globally. Students who learn it in school advocate for it professionally.

NGAME belongs in exactly this category. The precedent is well-established: auditing software that runs alongside the accounting system, not inside it, and that practitioners learn as students and carry into their careers.

Confirming the Career Toolbox Strategy

The educational market strategy you’re describing has a proven track record in this space. ACL and IDEA built significant market share specifically through university accounting programs — students who trained on them as undergraduates became the staff auditors and eventually partners who specified those tools on every engagement. NGAME targeting auditing students as early adopters is not just reasonable; it is the textbook playbook for this category.

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